

WASP-43b

R-1935

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190206 · created 2026-08-06T14:01:19+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458520.8285 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.056
Transit depth (Rp/Rs)^2	3.9 +/- 2.23 [%]
Orbital Inclination (inc)	86.16 +/- 2.86
Airmass coefficient 1 (a1)	1.45 +/- 0.38
Airmass coefficient 2 (a2)	-0.21 +/- 0.44
Scatter in the residuals of the lightcurve fit is	28.14 %
Best Comparison Star	#1 - [144, 29]
Optimal Aperture	2.49
Optimal Annulus	11.11
Transit Duration (day)	0.0599 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.056 vs 0.1594 (deviation 0.65σ)
Duration fit vs published	0.0599 d vs 0.0512 d (deviation 1.34σ)
Mid-time O–C	3.1 min
Depth SNR	3.3
Residual scatter	28.14 %

Light curve

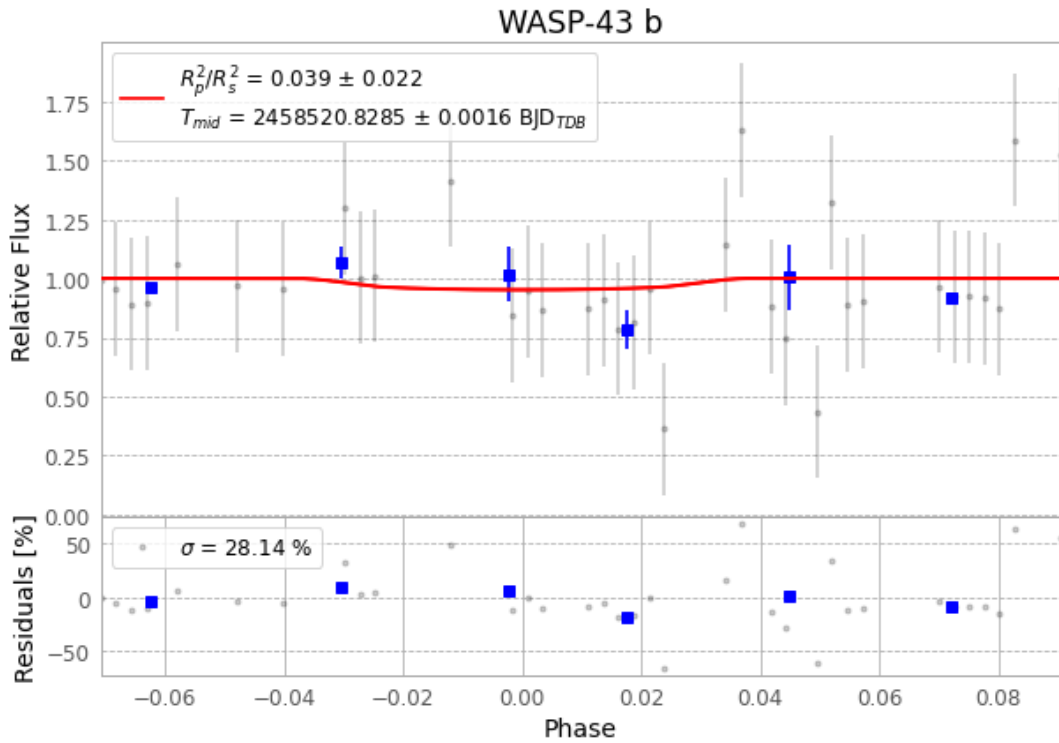


Figure 1 — FinalLightCurve_WASP-43 b_2019-02-05.png (SHA-256 c91831038d6ea915...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [440, 153], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 64afe2b58aa63ce8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143

Data provenance

64 FITS frames (42 MB), WASP-43190206062112.FITS → WASP-43190206093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190206.log (ccde085390c6...), FinalLightCurve_WASP-43 b_2019-02-05.png (c91831038d6e...), FinalLightCurve_WASP-43 b_2019-02-05.csv (6290166a0f58...)

Compute resources

Wall time 191.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:01Z.